

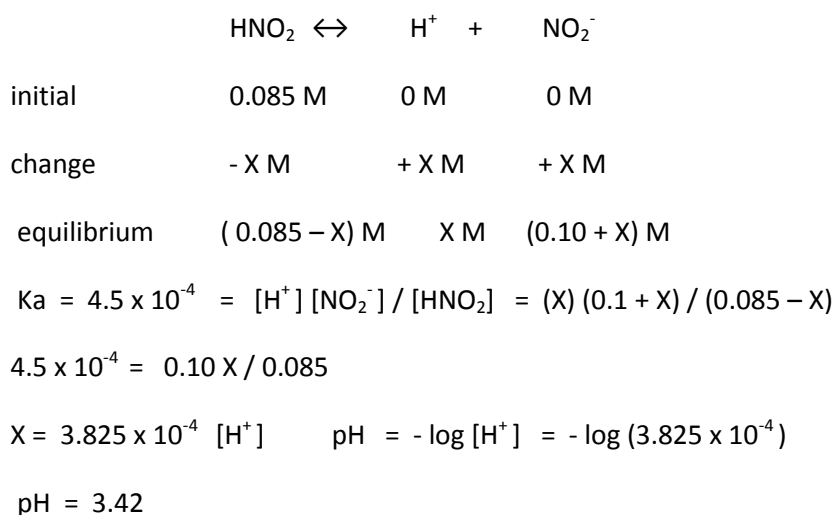
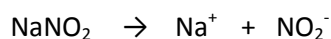
Chapter 17 exercises**Q1. Practice exercise page 721**

Calculate the pH of a solution containing 0.085 M nitrous acid HNO_2 ; $K_a = 4.5 \times 10^{-4}$) and 0.10 M potassium nitrite NaNO_2 .

Answer:

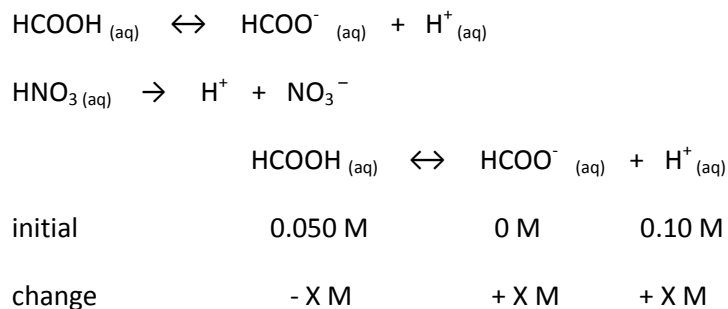
HNO_2 is a weak acid

NaNO_2 is a salt of strong electrolyte

**Q2. Practice exercise page 722**

Calculate the formate ion concentration and pH of a solution that is 0.050 M in formic acid (CHOOH) ($K_a = 1.8 \times 10^{-4}$) and 0.10 M HNO_3 .

Answer:



equilibrium (0.050 – X) X M (0.10 + X)

$$K_a = 1.8 \times 10^{-4} = [\text{HCOO}^-] [\text{H}^+] / [\text{HCOOH}] = (0.10 + X) (X) / (0.050 - X)$$

$$X = 0.9 \times 10^{-4} \text{ M } [\text{H}^+]$$

$$\text{Concentration of } [\text{H}^+] \text{ after equilibrium} = 0.10 + X = 0.10 + 0.9 \times 10^{-4} = 0.10 \text{ M}$$

$$\text{pH} = -\log (0.10) = 1.00$$

Q3. Practice exercise page 725

Calculate the pH of a buffer composed of 0.12 M benzoic acid and 0.20 M sodium benzoate.

$$(K_a = 6.3 \times 10^{-5})$$

Answer:

$$\text{pH} = \text{p}K_a + \log (\text{base}) / (\text{acid})$$

$$\text{p}K_a = -\log (6.3 \times 10^{-5}) = 4.2$$

$$\text{pH} = 4.20 + \log (0.2 / 0.12) = 4.20 + 0.22 = 4.42$$

Q4. Practice exercise page 728

Determine:

- The pH of the original buffer described in exercise 17.5 after the addition of 0.020 M HCl and
- The pH of solution that would result from the addition of 0.020 M HCl to 1.00 L of pure water.

Answer:

a)	$\text{CH}_3\text{COOH} \leftrightarrow \text{CH}_3\text{COO}^- + \text{H}^+$
initial	0.30 M 0.30 M 0 M
change	- 0 M 0.02 M
equilibrium	(0.30 + 0.02) M (0.30 – 0.02) M 0 M

$$\text{pH} = K_a + \log (0.28 / 0.32) = 4.68$$

$$\text{b) } \text{pH} = -\log (0.020) = 1.70$$

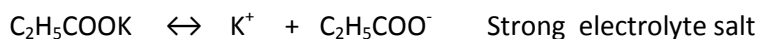
The small amount of HCl added change the pH to more acidic (1.70) of pure water, while the pH change of the buffer is very little.

17.15

Use the information in Appendix D to calculate the pH of:

- a) a solution that is 0.060 M in potassium propionate ($\text{C}_2\text{H}_5\text{COOK}$) and 0.085 M in propionic acid ($\text{C}_2\text{H}_5\text{COOH}$).
- b) a solution that is 0.075 M in tri methyl amine, $(\text{CH}_3)_3\text{N}$, and 0.10 M in tri methyl ammonium Chloride $(\text{CH}_3)_3\text{NHCl}$.
- c) a solution that is made of by mixing 50.0 mL of 0.15 M acetic acid and 50 mL of 0.20 M sodium acetate.

Answer:



initial	0.085 M	0 M	0 M
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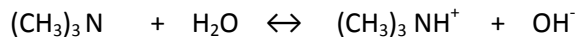
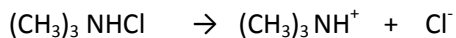
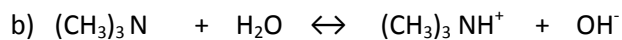
change	- X M	+ X M	+ X M
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equilibrium	$(0.085 - X)\text{M}$	X M	$(0.06 + X)\text{M}$
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$$K_a = 1.3 \times 10^{-5} = (X)(0.06 + X) / (0.085 - X) = 0.06 X / 0.085$$

$$X = 1.842 \times 10^{-5} \text{ M} = [\text{H}^+]$$

$$\text{pH} = -\log(1.842 \times 10^{-5}) = 4.73$$



initial	0.075 M	0.10 M	0 M
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change	- X M	+ X M	+ X M
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equilibrium	$(0.075 - X)\text{M}$	$(0.10 + X)\text{M}$	X M
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$$K_a = 6.4 \times 10^{-5} = (X)(0.10 + X) / (0.075 - X) = 0.10 X / 0.075$$

$$X = 4.8 \times 10^{-5} \text{ M} = [\text{OH}^-]$$

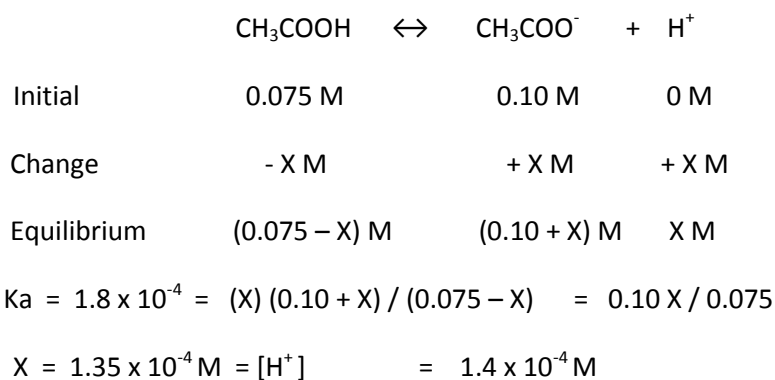
$$[\text{OH}^-][\text{H}^+] = 1.0 \times 10^{-14} \quad [\text{H}^+] = 1.0 \times 10^{-14} / 4.8 \times 10^{-5} = 0.21 \times 10^{-9} \text{ M}$$

$$\text{pH} = -\log(0.21 \times 10^{-9}) = 9.68$$

c) volume of solution = 50 mL + 50 mL = 100 mL

$$\text{mole of acetic acid} = (0.15 \text{ M} \times 50 \text{ mL}) / (100 \text{ mL}) = 0.075 \text{ M CH}_3\text{COOH}$$

$$\text{mole of sodium acetate} = (0.2 \text{ M} \times 50 \text{ mL}) / (100 \text{ mL}) = 0.10 \text{ M CH}_3\text{COONa}$$



17.17

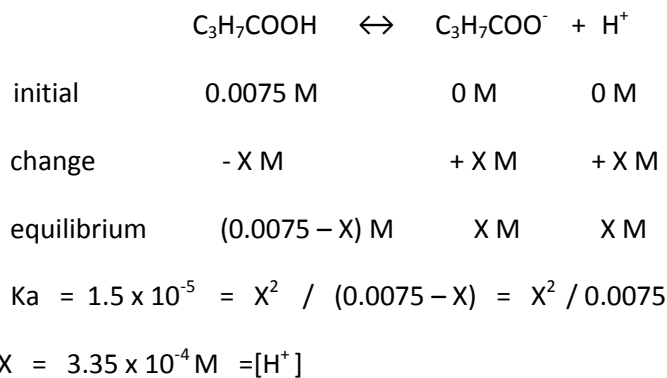
a) Calculate the percent ionization of 0.0075 M butanoic acid ($K_a = 1.5 \times 10^{-5}$).

b) Calculate the percent ionization of 0.0075 M butanoic acid in a solution containing 0.085 M sodium butanoate.

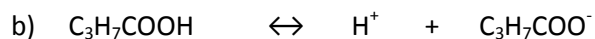
Answer:

a) $\text{C}_3\text{H}_7\text{COOH}$ butanoic acid is a weak acid

$\text{C}_3\text{H}_7\text{COONa}$ is a strong electrolyte salt



$$\text{Percent ionization} = \text{H}^+_{\text{equilibrium}} / [\text{HA}]_{\text{initial}} \times 100 = (3.35 \times 10^{-4}) / (0.0075) \times 100 = 4.5\%$$



initial	0.0075 M	0 M	0.085 M
change	- X M	+ X M	+ X M
equilibrium	(0.0075 - X) M	X M	(0.085 + X) M
$K_a = 1.5 \times 10^{-5} = (X)(0.085 + X) / (0.0075 - X) = 0.085 X / 0.0075$			
$X = 0.132 \times 10^{-5} \text{ M} = [\text{H}^+]$			
Percent ionization = $(0.132 \times 10^{-5}) \times 100 / (0.0075) = 0.018\%$			

17.21

- a) Calculate the pH of a buffer that is 0.12 M in lactic acid and 0.11 M in sodium acetate.
- b) Calculate the pH of a buffer formed by mixing 85 mL of 0.13 M lactic acid with 95 mL of 0.15 M sodium acetate ($K_a = 1.4 \times 10^{-4}$)

Answer:

a) $\text{pH} = \text{p}K_a + \log (\text{base} / \text{acid})$

$$\text{p}K_a = -\log K_a = -\log (1.4 \times 10^{-4}) = 3.85$$

$$\text{pH} = 3.85 + \log (0.11 / 0.12) = 3.85 + (-0.04) = 3.81$$

b) volume of solution = 85 mL + 95 mL = 180 mL

$$\text{concentration of lactic acid} = (0.13 \text{ M} \times 85 \text{ mL}) / (180 \text{ mL}) = 0.061 \text{ M}$$

$$\text{concentration of sodium lactate} = (0.15 \text{ M} \times 95 \text{ mL}) / (180 \text{ mL}) = 0.079 \text{ M}$$

$$\text{pH} = 3.85 + \log (0.079 / 0.061) = 3.96$$

17.25

How many moles of sodium hypobromate (NaBrO) should be added to 1.0 L of 0.050 M hypobromous acid (HBrO) to form a buffer solution of pH = 9.15? assume that no volume change occur when NaBrO is added. ($K_a = 2.5 \times 10^{-9}$)

Answer:

$$\text{p}K_a = -\log K_a = -\log (2.5 \times 10^{-9}) = 8.60$$

$$\text{pH} = \text{p}K_a + \log (\text{base} / \text{acid})$$

$$9.15 - 8.60 = \log (\text{base} / 0.050)$$

$$\text{Base} = [\text{NaBrO}] = 3.55 \times 0.05 = 0.177 \text{ M}$$

$$\text{mole of NaBrO} = 1.0 \text{ L} \times 0.177 \text{ mol/L} = 0.18 \text{ mole}$$

17.27

A buffer solution contains 0.10 mole of CH_3COOH and 0.13 mole of CH_3COONa in 1.0 L.

- what is the pH of this solution?
- What is the pH of the buffer after addition of 0.02 mole KOH?
- What is the pH of the buffer after addition of 0.02 mole of HNO_3 ?

Answer:

$$\text{a) } \text{pH} = \text{pK}_a + \log (\text{base/acid}) = 4.74 + \log (0.13 \text{ mol/L}) / (0.10 \text{ mol/L})$$

$$\text{pH} = 4.74 + 0.114 = 4.64$$

b)	CH_3COOH	+	OH^-	$\rightarrow$	H_2O	+	CH_3COO^-
buffer before addition	0.10 M		0 M				0.13 M
addition	-		0.02				-
buffer after addition	$(0.10 - 0.02) \text{ M}$		0 M				$(0.13 + 0.02) \text{ M}$

$$\text{pH} = \text{pK}_a + \log (\text{base} / \text{acid}) = 4.74 + \log (0.15 / 0.08) = 5.01$$

c)	CH_3COOH	$\leftrightarrow$	H^+	+	CH_3COO^-
buffer before addition	0.10 M		0 M		0.13 M
addition	0.020 M		0.02		0 M
buffer after addition	$(0.10 + 0.02) \text{ M}$		0 M		$(0.13 - 0.02) \text{ M}$

$$\text{pH} = \text{pK}_a + \log (\text{base} / \text{acid}) = 4.74 + \log (0.11 / 0.12) = 4.70$$

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